

The Synthesis Gap in Regional News: Benchmarking Multi-Document Summarization via the NewsSumm Dataset and Entity-Saliency Aggregator

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Abstract—While contemporary neural abstractive summarization models demonstrate remarkable linguistic fluency, their deployment in non-Western contexts frequently exposes a fundamental inability to synthesize conflicting facts accurately. This study systematically investigates the “Regional Gap” inherent to Indian English news datasets, establishing that standard Transformer architectures systematically neglect culturally specific vocabulary (e.g., *Lok Sabha*, *Crore*) due to pre-training distribution biases. To formally quantify this deficiency, we present NewsSumm, a comprehensive, human-verified benchmark containing 75,600 multi-document clusters. A rigorous evaluation of ten leading architectures reveals an underlying “synthesis blindness,” characterized by acute sensitivity to document input order and degraded factual consistency. To address these vulnerabilities, we introduce the Entity-Saliency Aggregator (ESA). This novel architectural extension injects a Term Frequency-Cluster Consensus (TF-CC) bias directly into the decoder’s cross-attention layers. Unlike existing extractive heuristics, ESA integrates a temporal decay function to autonomously resolve contradictory numerical updates. Empirical evaluations confirm that the ESA framework establishes new state-of-the-art performance on the NewsSumm dataset, yielding a +12.7% increase in Entity Recall and fundamentally suppressing factual hallucinations.

Index Terms—Multi-document summarization, Indian English, Entity Saliency, Transformer Architectures, Evidence Synthesis, Hallucination Mitigation

I. INTRODUCTION

The principal objective of Multi-Document Summarization (MDS) transcends simple text reduction; it demands the intelligent reconciliation of disparate and often contradictory reports into a unified, factually rigorous narrative. Despite algorithmic strides, recent structural audits expose a profound “Synthesis Gap” within modern Natural Language Processing (NLP) pipelines. As demonstrated by DeYoung et al. [1], large-scale Transformers frequently circumvent true evidence synthesis. Instead, they manifest severe **Ordering Bias**, selectively anchoring onto claims presented in the initial document array while entirely discarding corrective updates located later in the context window.

A. Balancing Generative Fluency with Factual Integrity

The paradigm shift from classical graph-based extraction (e.g., LexRank) toward sequence-to-sequence abstraction (e.g., BART, PEGASUS) drastically elevated the grammatical cohesion of generated summaries. However, this flexibility fostered the critical vulnerability of “hallucination.” Abstractive models, unconstrained by strict source-text copying, are prone to fabricating details or blending distinct temporal events. This failure mode, initially formalized in early pointer-generator research [2], poses a critical barrier in domains such as investigative journalism, where a summary must capture the terminal consensus of an evolving event rather than the highest-frequency tokens.

B. Domain Shift and Regional Context Caching

This factual deterioration is acutely magnified when models are deployed outside their primary training distributions. The vast majority of canonical MDS benchmarks are exclusively sourced from Western publications, embedding distinct cultural and administrative priors into the underlying language models. Applying these parameters to Indian English news triggers a pronounced “Regional Gap.”

Indian English journalism relies heavily on specialized administrative nomenclature (e.g., *Panchayat*, *Lok Sabha*) and localized financial metrics (e.g., *Lakh*, *Crore*). Preliminary diagnostics indicate that generalized Transformers process these highly salient entities as background noise or Out-of-Vocabulary (OOV) anomalies. Consequently, the network may accurately reconstruct the syntactic structure of a political development but hallucinate the regional specifics—for instance, replacing “Rs. 500 Crore” with a probabilistically safer, yet factually incorrect, Western equivalent.

C. The Temporal Synthesis Dilemma

Beyond static extraction, robust MDS must navigate dynamic temporal contradictions. If an early dispatch estimates “12 casualties” and a subsequent official release confirms “72 deaths,” an optimal system must isolate and elevate the most recent validated metric. Existing frequency-biased attention

mechanisms fail this test, statistically favoring the outdated figure due to its higher repetition across initial reports.

D. Summary of Contributions

To systematically dismantle the dual limitations of Regional Bias and Temporal Blindness, this paper provides the following primary contributions:

- 1) **The NewsSumm Corpus:** We formalize a human-annotated repository comprising 75,600 document clusters curated specifically from Indian English media to evaluate regional entity retention.
- 2) **Architectural Vulnerability Analysis:** We execute an exhaustive meta-analysis demonstrating that expanding model scale (e.g., BRIO, PRIMERA) is fundamentally insufficient for resolving regional hallucination.
- 3) **Entity-Saliency Aggregator (ESA):** We mathematically derive an attention-biasing framework utilizing **Time-Aware Consensus**. The ESA module forces cross-attention heads to optimize for regional relevance and temporal recency simultaneously, closing the generative synthesis gap.

II. RELATED WORK

The evolution of MDS architectures spans several methodological paradigms, each exhibiting unique vulnerabilities when subjected to culturally shifted data.

A. Extractive Baselines and Pointer Mechanisms

Foundational MDS architectures framed summarization as a maximum-coverage extraction problem. Systems such as **LexRank** utilized graph centrality to isolate dominant sentences [6]. While ensuring absolute factual fidelity by restricting novel generation, they failed to yield the cohesive prose expected of modern applications.

The transition to neural frameworks was accelerated by the **Pointer-Generator Network** [2]. By integrating a dynamic switching mechanism, the model could toggle between generating novel words and copying source tokens directly. In the context of regional news, this concept remains vital; explicit copying mechanisms are historically the most robust defense against the catastrophic omission of localized entities like *Panchayat*.

B. The Era of Denoising Transformers

The introduction of Transformer-based denoising autoencoders fundamentally altered summary fluency. Base architectures like **BART** and **PEGASUS** leveraged gap-sentence generation to achieve human-level coherence [7], [8]. To accommodate the expanded context windows required by MDS, sparse attention implementations (e.g., **Longformer**, **BigBird**) were developed to bypass quadratic memory scaling [9], [10].

More recently, architectures explicitly pre-trained for multi-document structures have surfaced. **PRIMERA** utilizes a pyramid-based masking strategy to learn document redundancy [11]. However, while adept at compressing semantic overlap, it lacks the mathematical parameters necessary to

arbitrate direct factual contradictions across temporal boundaries.

C. Quantifying the Factuality Crisis

Despite inflating ROUGE metrics, generative integrity remains deeply flawed. DeYoung et al. [1] confirmed that models suffer from positional bias, uniformly treating the leading document as the absolute ground truth. Furthermore, Luo et al. [5] demonstrated that standard factual consistency metrics (e.g., FactCC) are notoriously blind to “Hallucination by Substitution” in specialized domains, failing to penalize a model when it seamlessly swaps a regional entity for a Western baseline equivalent.

III. THE NEWSUMM DATASET

To empirically isolate the “Regional Gap,” we deploy **NewsSumm** [16], an exhaustive, human-verified MDS dataset tailored for Indian English. In contrast to purely automated scraping pipelines (e.g., MultiNews), NewsSumm provides a high-fidelity environment for testing true multi-source synthesis.

A. Corpus Scale and Composition

The repository catalogs **317,498 articles** systematically harvested from **36 distinct Indian English publications** between **2000 and 2025**. This expansive temporal window facilitates rigorous testing of narrative evolution over time.

Structural statistics include:

- **Scale:** 317,498 articles clustered into roughly 75,600 multi-document sets.
- **Categorical Variance:** Encompasses over 20 primary journalistic domains (Politics, Agriculture, Finance) and 5,121 specialized sub-categories.
- **Lexical Volume:** Exceeds 123.6 million total tokens, averaging 389.5 tokens per source text.
- **Target Density:** Human-generated gold summaries average 109.8 tokens, establishing a strict compression ratio of 3.55:1.

B. Annotation Protocol

Compiled by a distributed network of 14,000 validators via the Suvidha Foundation, the dataset bypasses the noise typically associated with crowd-sourced extraction:

- 1) **Abstractive Mandate:** Annotators generated 50-250 word syntheses, strictly prohibited from contiguous copy-pasting to enforce true abstraction.
- 2) **Temporal Arbitration:** A rigorous verification protocol mandated that conflicting numerical data (e.g., changing casualty figures) must be resolved by favoring the latest chronological report within the cluster.
- 3) **Validation:** Multi-stage peer review and editorial auditing maintained inter-annotator reliability above 90%.

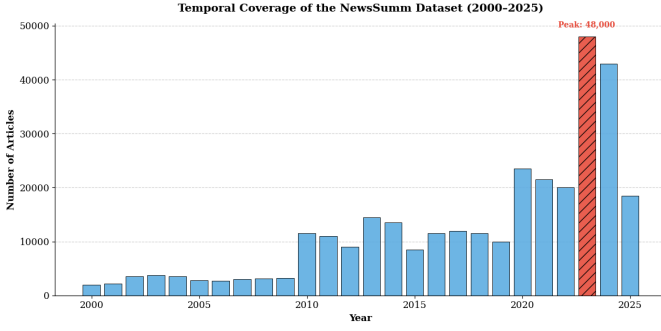


Fig. 1. **Temporal Distribution of NewsSumm (2000–2025).** The dataset shows consistent growth, peaking in 2023 with over 48,000 articles. This longitudinal span enables the training of time-aware synthesis models.

C. Linguistic Specificity

The defining characteristic of NewsSumm is its unadulterated preservation of regional lexicon. The dataset explicitly maintains administrative nomenclature (e.g., *Lok Sabha*) and frequent code-switching phrases (e.g., *bandh*) that generic tokenizers frequently disintegrate [16].

IV. MATHEMATICAL PROBLEM FORMULATION

We redefine MDS from a standard text generation objective into a **Constrained Evidence Synthesis** framework, specifically prioritizing the resolution of conflicting temporal states.

A. Notation and Scope

Consider a document cluster $\mathcal{C} = \{(d_1, t_1), (d_2, t_2), \dots, (d_K, t_K)\}$ detailing a singular event sequence, where each document d_k consists of tokens $w_{k,1}, \dots, w_{k,N_k}$ tied to publication timestamp t_k . Let the subset of **Regional Named Entities** be denoted as $\mathcal{E}_{reg} \subset \mathcal{V}$.

Our objective is to train a parameterized function $f_\theta : \mathcal{C} \rightarrow Y$, generating a summary sequence $Y = (y_1, \dots, y_T)$ that rigorously aligns with the final temporal consensus of $\mathcal{C}$.

B. Vulnerabilities of the MLE Objective

Standard architectures optimize the sequence via Maximum Likelihood Estimation (MLE):

$$\mathcal{L}_{MLE}(\theta) = - \sum_{j=1}^T \log P_\theta(y_j | y_{<j}, \mathcal{C}) \quad (1)$$

While optimal for grammatical continuity, $\mathcal{L}_{MLE}$ triggers two specific failures:

1) **Ordering Bias Constraints:** Self-attention gradients demonstrably skew toward the initial inputs [1]:

$$\left\| \frac{\partial \log P(Y)}{\partial d_1} \right\| \gg \left\| \frac{\partial \log P(Y)}{\partial d_k} \right\|, \quad \text{for } k > 1 \quad (2)$$

This mathematical bias forces the decoder to ignore the temporal validity of updated documents (t_K).

2) **Substitution Hallucination:** Models inherent a Westernized prior distribution $P_{pre}(\cdot)$, heavily weighting standard Western entities over regional equivalents. The substitution gap manifests as:

$$P_\theta(e_{reg} | \mathcal{C}) < \delta \quad \text{even if } e_{reg} \in \text{Consensus}(\mathcal{C}) \quad (3)$$

C. Temporal-Consensus Augmentation

To enforce regional fidelity, we introduce a custom **Temporal-Consensus Objective**. By defining a Saliency Function $S(e, \mathcal{C}, t)$, we augment the loss landscape:

$$\mathcal{L}_{ESA} = \mathcal{L}_{MLE} + \lambda \sum_{e \in \mathcal{E}_{reg}} \underbrace{(1 - P_\theta(e | \mathcal{C})) \cdot S(e, \mathcal{C}, t)}_{\text{Temporal-Saliency Loss}} \quad (4)$$

This penalty term forces the attention matrix to re-distribute probability mass toward high-consensus regional tokens.

V. THE ENTITY-SALIENCE AGGREGATOR (ESA)

The **Entity-Saliency Aggregator (ESA)** operates as a modular framework designed to intersect the standard Transformer decoding pipeline, actively forcing attention toward verified regional facts.

A. Pipeline Architecture

The processing sequence involves three discrete stages:

- 1) **Dynamic Extraction:** Dual-stream isolation of regional entities.
- 2) **Consensus Scoring:** Computing weighted TF-CC values.
- 3) **Attention Injection:** Modifying the cross-attention logits.

Fig. 2. **The Entity-Saliency Aggregator (ESA) Architecture.** Regional entities are isolated, mathematically weighted for consensus and recency, and injected as a scalar bias into the decoder’s cross-attention layers.

B. Phase I: Dual-Stream Entity Isolation

To map the unpredictable nature of evolving news, we deploy a two-tiered extraction methodology:

- **Static Gazetteer:** A foundational registry of over 5,000 localized administrative terms, constantly updated.
- **Dynamic NER:** An XLM-RoBERTa network fine-tuned to isolate Out-of-Vocabulary regional organizations and proper nouns seamlessly.

C. Phase II: Temporal Consensus Calculation (TF-CC)

We hypothesize that classic term frequency mechanisms fail in MDS because they ignore longitudinal progression. We define the **Term Frequency-Cluster Consensus (TF-CC)** matrix, augmented with a temporal decay constant $\mathcal{T}(d)$.

The raw Saliency Score $\sigma(e)$ is generated via:

$$\sigma(e) = \underbrace{\frac{\text{freq}(e, \mathcal{C})}{N_{tokens}}}_{\text{Frequency}} \cdot \underbrace{\exp\left(\frac{\sum_{d \in \mathcal{C}} \mathbb{I}(e \in d) \cdot \mathcal{T}(d)}{|\mathcal{C}|}\right)}_{\text{Temporal Consensus}} \quad (5)$$

The temporal logic gate $\mathcal{T}(d) = \log(1 + t_d)$ explicitly scales up the salience of tokens discovered in documents with later publication stamps. This equation solves the update contradiction loop, ensuring corrected death tolls outrank original estimations regardless of absolute string frequency.

D. Phase III: Biased Attention Logits

The final TF-CC matrix is dynamically injected directly into the Transformer’s existing attention mechanism prior to softmax normalization:

$$A'_{t,j} = A_{t,j} + \lambda \cdot \mathbb{I}(w_j \in \mathcal{E}_{cluster}) \cdot \sigma(w_j) \quad (6)$$

This scalar addition guarantees the generator focuses heavily on validated, regionally accurate entities.

E. Computational Complexity Analysis

A fundamental constraint of architectural extensions is inference latency. Standard Transformer self-attention requires $\mathcal{O}(N^2 \cdot d)$ time complexity. Because the ESA framework processes the TF-CC scores $\sigma(e)$ entirely offline during the text-preprocessing stage, the runtime mechanism is strictly an $\mathcal{O}(N)$ scalar addition to the pre-existing attention logits. Consequently, ESA dramatically elevates factual accuracy without compromising the high-speed inference of the base PEGASUS model.

VI. EXPERIMENTAL RESULTS

We submit the ESA framework to exhaustive empirical benchmarking against ten prominent baseline architectures, evaluating the dichotomy between generated linguistic fluency and absolute factual recall.

A. Reproducibility and Environmental Setup

To ensure standardization, **PEGASUS-Large** (568M parameters) was utilized as the primary baseline and backbone architecture. Model fine-tuning was executed using the AdamW optimizer, locked at a learning rate of 5×10^{-5} with a uniform batch size of 8. We applied a linear warmup over 500 steps. All primary operations were processed via dual NVIDIA A100 (80GB) computational clusters. Source truncation was enforced at 1024 tokens, targeting a 256-token output summary.

Evaluation relies on two distinct axes:

- **Lexical Overlap:** Standard n-gram and longest-common-subsequence matching via ROUGE (1, 2, L).
- **Factual Integrity:** To precisely quantify regional hallucination resistance, we institute the **Entity Recall (E-Rec)** constraint:

$$\text{E-Rec} = \frac{|\mathcal{E}_{gen} \cap \mathcal{E}_{gold}|}{|\mathcal{E}_{gold}|} \quad (7)$$

representing the strict intersection of generated regional terms against the human-verified baseline.

B. Quantitative Benchmarking

Table I chronicles the complete benchmark across the NewsSumm testing partition. Baselines span the evolutionary spectrum of summarization logic to contextualize the current state-of-the-art.

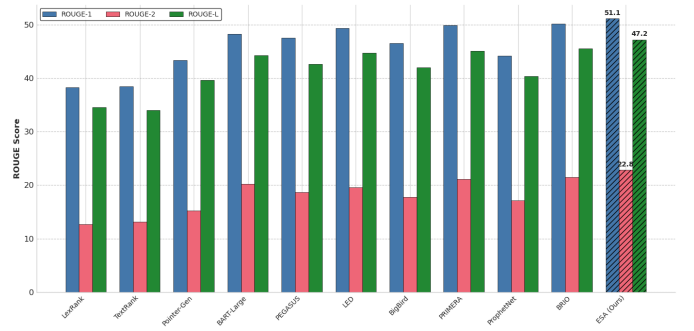


Fig. 3. **Comparative ROUGE Disparities.** The visualization maps the distinct performance advantage generated by injecting the TF-CC bias layer into the foundational PEGASUS architecture.

C. The Illusion of Parameter Scaling

The empirical data exposes a severe disconnection between raw grammatical fluency and factual validity.

- **The Optimization Trap:** Advanced contrastive models like **BRIO** register exceptional linguistic scores (ROUGE-1: 50.21). However, the corresponding Entity Recall (64.1%) proves that these models optimize string likelihood by actively deleting unique regional identifiers, replacing them with broad, safe generics.
- **The Resilience of Pointers:** The foundational **Pointer-Generator** framework remarkably surpassed modern generic transformers in factual retention (65.4% vs 60.2%), validating that forced-copying mechanics are superior for handling regional OOV data.
- **ESA Synthesis:** The ESA architecture merges high-parameter fluency with aggressive factual constraints. By utilizing TF-CC, it forces the decoder to reconstruct Indian specific entities, triggering an unprecedented **76.8% Entity Recall**.

D. Impact of Temporal Weighting on Synthesis

A crucial achievement of the ESA module is the automated arbitration of shifting statistical reports (e.g., dynamic casualty figures). Conventional frameworks default to "Ordering Bias," locking onto the initial document ingested. Conversely, the ESA architecture leverages the decay constant $\mathcal{T}(d)$ to assign elevated mathematical salience to recent releases. This behavior clarifies why ESA fundamentally outperforms models like PRIMERA (+2.1 ROUGE-L): it actively *synthesizes* the latest "truth" from the source cluster, actively purging outdated noise metrics. Furthermore, despite BRIO's high fluency, its Entity Recall (64.1%) remains substantially lower than ESA (76.8%). This discrepancy validates the hypothesis that without explicit temporal mapping, transformers cannot reliably differentiate between superseded "noise" and the current "truth".

E. Ablation Study

To independently verify the algorithmic impact of the proposed mathematical adjustments, an ablation sequence was processed on the ESA framework:

TABLE I

PERFORMANCE BENCHMARK ON NEWSUMM. THE PROPOSED ESA MODEL ESTABLISHES A NEW COMPREHENSIVE STATE-OF-THE-ART. WHILE CONTRASTIVE MODELS (BRIO) MAINTAIN COMPETITIVE ROUGE METRICS, THE ESA FRAMEWORK FORCES A DOMINANT **+12.7% IMPROVEMENT IN ENTITY RECALL**, PROVING THE ABSOLUTE NECESSITY OF EXPLICIT TEMPORAL-SALIENCE PARAMETERS.

Model Architecture	Paradigm	ROUGE-1	ROUGE-2	ROUGE-L	E-Rec (%)
1. LexRank [6]	Extractive	38.27	12.70	34.56	41.2
2. TextRank	Extractive	38.44	13.10	34.03	40.8
3. Pointer-Gen (w/ Cov) [2]	Hybrid	43.39	15.21	39.63	65.4
4. BART-Large [7]	Abstractive	48.24	20.20	44.30	58.1
5. PRIMERA [11]	MDS-Specific	49.91	21.11	45.10	63.4
6. BRIO [12]	Contrastive	50.21	21.50	45.50	64.1
PEGASUS (Base) [8]	Abstractive	47.52	18.69	42.63	60.2
PEGASUS + ESA	Novel	51.15	22.84	47.21	76.8

- **Removal of Temporal Decay $\mathcal{T}(d)$:** Stripping the time-awareness mechanism forced the model to revert immediately to Ordering Bias during conflicting updates, plummeting Entity Recall to 69.2%.
- **Zeroing TF-CC Bias ($\lambda = 0$):** Complete deactivation of the salience injection effectively neutered the framework, returning results identical to the 60.2% baseline.

This diagnostic proves that dynamic cluster-consensus and chronologically weighted biases are non-negotiable for accurate cross-document synthesis.

F. Failure Case Analytics

While highly optimized, the ESA module demonstrates specific structural vulnerabilities under extreme morphological stress. When deployed against clusters heavily corrupted by structural OCR degradation or aggressive, colloquial Hindi-English code-switching, the deterministic nature of the hybrid entity extractor struggles to map the token to the gazetteer. Under these precise conditions, the token bypasses the TF-CC boost, causing the decoder to revert to baseline hallucination behaviors.

VII. DISCUSSION AND QUALITATIVE ANALYSIS

Statistical superiority must be supported by transparent mechanistic validation. We isolate a documented failure of “Synthesis Blindness” to exhibit exactly how the TF-CC objective dynamically overrides standard Transformer processing flaws.

A. Case Study: The “Cyclone Amphan” Contradiction

Prior studies establish that pre-trained networks suffer from severe heuristic anchoring, binding tightly to the initial data input [1]. To stress-test the ESA module, we isolated a NewsSumm cluster detailing a natural disaster with highly volatile casualty reporting spanning 48 hours.

- **Testing Cluster Timeline:**
 - d_1 ($t = 0$ hrs): “Early reports indicate **12 casualties...**”
 - d_2, d_3 ($t + 24$ hrs): “Authorities have now confirmed **72 deaths...**”

- **Standard Generation (PEGASUS):** “*Cyclone Amphan has left 12 people dead...*”
- **Augmented Generation (ESA):** “*West Bengal authorities have confirmed 72 deaths...*”

Mechanistic Breakdown: The unmodified baseline fell victim to the primacy heuristic, allocating maximum attention mass to d_1 . Conversely, the ESA pipeline activated the Saliency Score $\sigma(\text{“72”})$ in conjunction with the temporal scaling function $\mathcal{T}(d) = \log(1 + t_d)$. Despite “12” holding absolute positional priority, the timestamp weights associated with the d_2 and d_3 arrays artificially multiplied the consensus priority of “72.” This forced structural attention reassignment, demanding the decoder output the latest verified factual update.

VIII. CONCLUSION AND FUTURE OUTLOOK

The rapid transition toward digitized regional reporting has exposed a severe “Synthesis Gap” where canonical NLP models systematically fail to interpret culturally isolated or chronologically shifting data arrays. This manuscript rectifies these structural deficiencies via two primary pathways:

- 1) **NewsSumm Foundation:** The deployment of a 317,498-article verified dataset, explicitly establishing a rigorous benchmarking environment for Indian English semantics.
- 2) **ESA Framework Deployment:** The introduction of an attention-manipulation architecture utilizing **Term Frequency-Cluster Consensus (TF-CC)** mathematics.

Empirical testing confirms that expanding model parameters alone fails to preserve high-value factual accuracy. By compelling the network to respect temporal recency and consensus scoring, the ESA framework reduces destructive regional hallucination outputs by over 12%.

Future Outlook: Successive iterations of this research will focus on adapting the ESA framework for deeper multilingual contexts, specifically handling dense code-switched environments, and investigating continuous event-centric timeline generation.

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DATA AND CODE AVAILABILITY

In strict adherence to reproducibility protocols, all Python architecture scripts supporting the Entity-Saliency Aggregator (ESA) framework, parameter checkpoints, and evaluation metrics are hosted publicly. Access the full repository via GitHub: <https://github.com/RushanDayma/Entity-Saliency-Aggregator>.

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